

What is the process once a new national mission is conceptualised under the PSA's office?

These missions aim to make India future-ready in critical and emerging technologies while ensuring that the benefits of innovation reach end users across the country. All stakeholders are engaged throughout the process, though implementation rests with the respective ministries.

Once a mission is conceptualised and approved, it is developed in close collaboration with the relevant ministry for implementation. This inclusive approach enables academia, industry, startups, and government bodies to contribute their expertise and collectively advance technology, innovation, and national capacity-building.

What is your view on the rare-earth metals challenge that India faces?

India is actively addressing the challenge of rare earth elements with the goal of becoming self-sufficient, or *Aatmanirbhar*, in this critical area.

Rare earth elements are not truly rare, but their extraction is complex and technology-intensive. In India, this work is led by India Rare Earths Limited (IREL) under the Department of Atomic Energy, which extracts rare earth oxides from monazite sands, a process that has been ongoing for many years. Earlier, the focus was on thorium and uranium, but rare earths have now become equally important.

IREL has developed the capability to extract both light and heavy rare earth oxides, including those used in permanent magnets such as neodymium and praseodymium. India currently produces about 1500 tonnes of rare earth oxides annually.

The next step involves converting these oxides into metals and alloying them with iron and boron to produce rare earth magnets. This entire technology chain has been developed domestically, and several

Indian companies are preparing to manufacture these magnets using domestically sourced materials. While this marks a significant achievement, production capacity must expand to meet growing demand from electric vehicles, wind energy, and electronics.

At the same time, India is investing in rare-earth-free permanent magnets that achieve high magnetisation without using rare earth elements. Some Indian companies already produce these magnets, which, although slightly larger, are cost-effective and reduce dependence on imports.

Overall, India's efforts are well underway to achieve technological self-reliance and global competitiveness in this strategically vital field.

Let us start with one of your major initiatives. Which one would you like to discuss first, and what makes it particularly significant?

I would like to shed light on the **Manthan initiative**, launched on 15 August 2022. It is a platform that connects research, academia, industry, startups, and government on a common stage. Manthan bridges those presenting technology-related problem statements with those capable of solving them, including researchers, startups, and companies. Beyond hosting problem statements online, the platform involves active consultation and guidance to ensure meaningful and impactful collaborations.

In just over three years, **projects worth approximately ₹87 billion have been initiated, with over ₹15 billion transacted**, much of it directed to startups and academic institutions. The platform has also led to the creation of several Centres of Excellence.

At its core, Manthan seeks to unite the ecosystem to address societal challenges and has even drawn international participation, fostering global innovation for the public good.

So who exactly can use the Manthan platform, and how can a common person make use of it?

To start using Manthan, participants must register on the platform to verify their authenticity. Users can then participate either on the supply side or the demand side.

The supply side includes those offering solutions, such as academic and research institutions, startups, grassroots innovators, and private organisations with innovative technologies or ideas. The demand side includes those seeking solutions, such as large corporations, PSUs, government departments, state agencies, philanthropists, and foundations working in areas such as healthcare, waste management, and rural development.

All participants are verified to ensure authenticity and prevent misuse. The Manthan team facilitates collaboration by connecting partners and supporting them through a transparent, well-managed process that safeguards privacy and integrity.

Once an idea is submitted on the Manthan platform, what kind of support is provided, such as funding, mentorship, or validation?

Manthan does not provide direct funding, fellowships, or financial support. Its primary role is to connect those facing societal challenges with individuals or organisations that have the capability or technology to address them.

Once a collaboration is established, the participating parties determine the terms, including deliverables and funding arrangements. So far, projects worth over ₹15 billion have been executed through such partnerships, demonstrating that real transactions and implementation take place through direct agreements between participants.

Is there any initiative that focuses on Tier 2 and Tier 3 colleges in India?